QA Automation & Performance

Senior Quality Engineer – Automation & Performance Assessment

Introduction

This assessment is designed to evaluate the skills of a **Senior Quality Engineer** with strong experience in **test automation (Playwright)** and **performance testing**.

We are not only interested in whether things work, but **how you think**:

- Test design and structure
- Reliability and maintainability
- Performance reasoning and interpretation
- Engineering trade-offs and clarity of communication

The assessment consists of **two practical tasks** and a **short reflection section**.

Expected time investment: ~3–4 hours
Tools: Playwright + one performance testing tool of your choice

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Practical Assignment

Task 1 – UI Automation with Playwright

Target application:
https://www.saucedemo.com

Requirements

Implement automated UI tests using **Playwright** that cover the following scenarios:

- Successful login using:
  - Username: `standard_user`
- Failed login using:
  - Username: `locked_out_user`
- Add a product to the cartw
- Start the checkout process

Technical Expectations

- Use **Playwright Test** with a clean project structure
- Use **fixtures** or reusable setup logic
- Use **explicit waits and Playwright best practices**
  - No hardcoded sleeps (`sleep` , `waitForTimeout`)
- Ensure tests are:
  - Deterministic
  - Readable
  - Maintainable
- Provide a **CLI command** to run the tests locally
- Include a **README** explaining:
  - How to run the tests
  - Any design decisions you made

Optional (Bonus)

- Configure Playwright to run in **GitHub Actions**
- Add:
  - HTML test reports
  - Screenshots or traces on failure

Evaluation Criteria

- Test structure and readability
- Correct use of Playwright APIs
- Reliability and stability of tests
- Reusability and maintainability
- Quality of documentation

Task 2 – Performance Testing

Target API:

https://reqres.in/api/users?page=1

Tool (choose one):

- k6 (preferred)
- JMeter
- Artillery

Scenario

- Simulate **100 concurrent users**
- Each user sends **1 request per second**
- Test duration: reasonable to capture stable metrics (explain your choice)

Metrics to Capture

- Response time:
  - P50
  - P95
  - P99
- Error rate
- Throughput (requests/sec)

Deliverables

- Performance test script
- Short performance report including:
  - Test setup and assumptions
  - Captured metrics
  - Interpretation of results
  - Potential optimization suggestions (even if the system is simple)

Evaluation Criteria

- Correct test setup and execution
- Proper metrics collection
- Ability to interpret results
- Clarity and realism of conclusions
- Understanding of performance testing principles

Reflection & Seniority Check

Please answer briefly in the README or a separate document:

- How would you integrate **Playwright tests into CI/CD**?
- How would you **notify the team (Slack/Teams)** about failures or regressions?
- What **observability metrics** would you include in an end-to-end quality dashboard?
- How would you decide:
  - What to automate
  - What not to automate

- What belongs to performance vs functional testing

(Short, pragmatic answers are sufficient.)

Submission Instructions

- Submit via:
  - GitHub repository **(preferred)**
  - or ZIP archive
- Include:
  - Source code
  - README with setup and explanations
  - Performance report (Markdown or PDF)

What We Are Looking For

This is not about perfection.

We value:

- Clean, pragmatic engineering
- Good testing instincts
- Clear communication
- Senior-level reasoning and trade-offs